

TABLE 3—EFFECT OF THE REFORM, SEPARATELY FOR EACH MIDDLE SCHOOL GPA QUINTILE

|                               | Quintile 1        | Quintile 2        | Quintile 3        | Quintile 4        | Quintile 5        |
|-------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Competitive school enrollment | −0.246<br>(0.036) | 0.066<br>(0.047)  | 0.052<br>(0.042)  | 0.151<br>(0.047)  | 0.210<br>(0.049)  |
| High school completion        | −0.077<br>(0.041) | −0.003<br>(0.029) | 0.004<br>(0.022)  | 0.031<br>(0.020)  | −0.006<br>(0.018) |
| University enrollment         | −0.066<br>(0.042) | 0.022<br>(0.037)  | −0.035<br>(0.027) | −0.004<br>(0.029) | −0.031<br>(0.024) |
| University completion         | −0.068<br>(0.050) | −0.018<br>(0.038) | −0.008<br>(0.033) | 0.005<br>(0.032)  | −0.020<br>(0.021) |

Notes: Estimates of parameters of an RDD regression where students’ outcome is regressed on a dummy for being born after December 31, 1988 (thus affected by the reform), and a linear trend in week of birth, allowed to be different for cohorts affected and not affected by the reform (with weeks of birth normalized to 0 on the first week of 1989). These control variables are interacted with dummies for the student belonging to each of the five quintiles of the middle school GPA distribution in the respective cohort. The regressions also include controls for seasonal dummies for the week of birth within the year (52 dummies). SE clustered at the week of birth level. The reported numbers are the estimates and the SE of the parameters on the interaction between the dummy for being born after December 31, 1988, and the dummies for each quintile, as indicated in the top row.

within-Bergen RDD framework and interacting the aggregate RDD effect with dummies for middle school GPA quintiles.<sup>40</sup> As expected, the proportion of students within the first quintile of middle school GPA distribution who attend a competitive high school decreases substantially as a consequence of the reform, while the same proportion increases for students with high grades (top two quintiles). The pattern of the effect on the three outcomes (high school completion, university enrollment, and university completion) is similar to what we observed when comparing Bergen with other municipalities. It is generally negative for the first quintile and still negative, but of smaller magnitude, for the top quintile.

Both cross-market comparison and within-Bergen analysis confirm that, if anything, the reform had a small negative aggregate impact. The estimated overall impact of the reform is the reduced form counterpart of the LATE when competitive high school attendance is instrumented by the reform—that is, the numerator of *LATE* equation in Section III. As mentioned, because this parameter does not separate between the effect for CCs and CPs, it can hide significant heterogeneity, as the analysis by quintile of GPA seems to indicate. The rest of this section exploits the econometric framework described in Sections III and IV to isolate the effect for the different types of compliers and more directly relate the observed changes in outcome to the changes in high school allocation caused by the reform.

B. Treatment Effects for CPs and CCs

Panel A in Table 4 shows the estimate of the first-stage parameters (equation (6)). In particular, column 1 reports the estimate of  $b_{CC}^{Below}$  and  $b_{CP}^{Below}$  from the first stage of the endogenous variable  $D_i \times Below_i$ —that is, the interaction between the dummy for competitive school attendance and the one for being below the cutoff. Column 2

<sup>40</sup> See equation (E.4) in Supplemental Appendix E for details.